

Adepu Nikhil

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SUMMARY

M.Sc. Artificial Intelligence and Data Science graduate with a B.Sc. in Electronics and hands-on experience developing AI-powered applications using Python, machine learning, computer vision, Generative AI, FastAPI, and embedded systems. Experienced in developing real-time AI solutions, integrating machine-learning models with software and hardware systems, and performing implementation, testing, debugging, and documentation. Published research on real-time assistive navigation using spatial reasoning and computer vision. Strong interest in AI engineering, Generative AI, computer vision, intelligent systems, research, and technology development.

ACADEMIC BACKGROUND

Master of Science, Artificial Intelligence and Data Science | Central University of Andhra Pradesh Sep 2024 – May 2026

Master's Dissertation: Distance-Aware Real-Time Assistive Navigation System Using YOLOv8 and Monocular Depth Estimation

- Developed a spatial guidance framework combining YOLOv8 real-time object detection with monocular geometric depth estimation to provide directional awareness (left/center/right) for visually impaired users.
- Implemented scalable FastAPI microservices and multilingual audio feedback, validating latency and accuracy under real-time constraints.

B.Sc. (Hons.), Electronics | University of Delhi, New Delhi

Sep 2021 – May 2024

RESEARCH & PROFESSIONAL EXPERIENCE

AI Intern

Infosys Springboard | Bangalore, India (Remote)

Aug 2025 – Oct 2025

Project Title: AI-Powered Distance-Aware Navigation System for the Visually Impaired

- Developed a real-time AI application to assist visually impaired users through object detection, distance estimation, spatial awareness, and audio feedback.
- Implemented **YOLOv8-based object detection** for identifying objects from camera input.
- Developed monocular vision-based distance estimation using detected object bounding-box information.
- Classified detected objects according to their spatial position as **left, center, or right**.
- Prepared and managed datasets using **Roboflow** for object detection model development.
- Integrated FastAPI for backend processing and communication between system components.
- Implemented multilingual audio feedback supporting **English, Hindi, and Telugu**.
- Tested and refined the system for real-time operation and environmental understanding.

Data Science Intern

Codtech IT Solutions Pvt. Ltd | Hyderabad, India (Remote)

May 2025 – Jul 2025

Project Title: AI-Powered Intelligent Loan Approval and EMI Analysis Platform

- Developed an **AI-powered Intelligent Loan Approval and EMI Analysis Platform** using Python, machine learning, and FastAPI.
- Built REST APIs to integrate machine-learning functionality with the application.
- Performed data preprocessing, feature handling, and analysis using Pandas and Scikit-learn.
- Applied machine-learning techniques to support automated financial decision analysis.
- Converted machine-learning functionality into an application-oriented backend system.
- Tested and validated API functionality and model-driven application workflows.

SKILLS

Programming:	Python, C++, C, MySQL
AI / ML & Vision:	Machine Learning, Deep Learning, Computer Vision, YOLOv8, OpenCV, Roboflow, Scikit-learn
Generative AI:	Google Gemini API, Ollama, Llama-based Local Models, Prompt Engineering, NLP
Backend & Web:	FastAPI, REST APIs
Embedded Systems:	ESP32, Arduino Uno, Embedded C/C++, Hardware-Software Integration, Microcontrollers
Tools & Cloud:	Git, GitHub, Docker, Jupyter Notebook, AWS EC2, Azure, MySQL, SQLite
Languages:	English, Hindi, Telugu

PROJECTS

AI-Integrated Embedded Automation and Monitoring System

- Developed an AI-powered assistive mobility system using **YOLO-based real-time object detection** to identify surrounding objects.
- Integrated the AI system with an **ESP32-based embedded platform** for real-time operation.
- Connected the ESP32 system with a smartphone through **Wi-Fi communication** for transferring system information and feedback.
- Implemented **real-time audio feedback** to communicate detected objects, spatial direction, and navigation-related information.
- Developed the processing pipeline: **camera input → AI object detection → spatial classification → ESP32 → Wi-Fi → smartphone → audio feedback**.
- Implemented spatial classification of detected objects as **left, center, or right**.
- Worked with **Python, YOLO, OpenCV, Roboflow, ESP32, Wi-Fi communication, and real-time AI processing**.
- Performed system integration, debugging, testing, and validation across AI, embedded, and communication components.

Offline AI Second Brain / NLP Conversational Assistant

- Developed an offline AI assistant concept using locally hosted language models for private, locally processed AI workflows.
- Worked with **Ollama and Llama-based models** for local model inference and AI-assisted information processing.
- Explored prompt engineering and structured interaction with locally deployed language models.
- Investigated approaches for building AI functionality without depending entirely on external cloud-based inference.
- Worked with local AI tools to understand practical deployment and integration of Generative AI systems.

Real-Time Voice Deepfake Detection System

- Developed a deep learning-based voice deepfake detection system to distinguish between genuine human speech and AI-generated or manipulated speech.
- Designed an audio preprocessing pipeline for speech loading, normalization, preprocessing, and feature preparation for deep learning.
- Extracted Mel-spectrogram features using Librosa to capture frequency and temporal characteristics of speech signals.
- Developed a CNN-based classifier using PyTorch to learn patterns from Mel-spectrograms and classify audio as real or deepfake.
- Implemented the complete workflow including dataset preparation, train-validation splitting, model training, loss monitoring, and evaluation.
- Evaluated the system using accuracy, precision, recall, F1-score, and confusion matrix, analyzing false positives and false negatives.

PUBLICATIONS

A Real-Time Assistive Navigation Framework for Visually Impaired Users Using Spatial Reasoning and Computer Vision

International Journal of Creative and Open Research in Engineering and Management (IJCOPE) | May 2026

DOI: 10.55041/ijcope.v2i5.171

- Developed and documented a real-time assistive navigation framework combining computer vision and spatial reasoning for visually impaired users.
- Explored real-time object detection and spatial understanding to identify surrounding objects and determine their relative positions.
- Investigated distance-aware and directional information for generating actionable navigation feedback.
- Studied the application of intelligent visual perception for real-world assistive navigation.

RESEARCH / DEVELOPMENT EXPERIENCE

- Experience developing AI solutions from **problem definition and data preparation through implementation, testing, debugging, and documentation.**
- Hands-on experience integrating AI models with backend services and embedded hardware.
- Experience working with **Generative AI and locally hosted language models.**
- Experience documenting technical implementations, project workflows, experiments, and system results.
- Experience using Git/GitHub for software development and project version control.

CERTIFICATIONS

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| • Fundamentals of Object Oriented Programming, NPTEL | Apr '25 |
| • Artificial Intelligence Primer Certification, Infosys Springboard | Jun '25 |
| • Agentic AI Project-Based Learning Program, IBM Skills Build (CSRBOX) | Aug '25 |
| • Principles of Generative AI Certification, Infosys Springboard | Jun '25 |
| • Quantum Fundamentals Program (4 Weeks) – WISER & Amaravati Quantum Valley | Feb '26 |

WORKSHOPS

- One-Day Workshop: 'Accelerate Your AI Journey: Machine Learning with Python' (Central University of Andhra Pradesh - Dept. of CS & AI)
- Two-Day Workshop: 'Design Thinking for Innovation: A Hands-on Approach to Problem Solving' (Central University of Andhra Pradesh – Dept. of CS & AI)